Artificial Intelligence in Laparoscopic Skill Assessment: A Scoping Review.

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Abstract

The purpose of this study was to describe and synthesize the existing literature regarding the application of artificial intelligence (AI) for laparoscopic skill assessment in surgical education. This scoping review was conducted in accordance with the PRISMA-ScR checklist. Literature search was performed across the MEDLINE, EMBASE, and Web of Science databases to identify original research published between 2015 and May 16, 2024. Eligible sources included studies evaluating AI-based assessments involving medical students, surgical residents, or fellows. Screening and full-text review were performed independently and in duplicate using Covidence, with discrepancies resolved by consensus. Academic and clinical training environments. Of 1973 abstracts screened, 30 original research studies met the final inclusion criteria. The participants were medical students, surgical residents, and fellows undergoing technical proficiency training in laparoscopic procedures. A total of 1973 abstracts were identified, and 30 studies were included for analysis. These were categorized into AI versus traditional/manual scoring (n = 12, 40%) and AI for distinguishing surgical skill levels (n = 18, 60%). Most studies assessed combinations of traditional laparoscopic skills of peg transfer (n = 9), suturing (n = 7), knot tying (n = 6), or pattern cutting (n = 6). AI models demonstrated moderate to high accuracy in differentiating novice and expert surgeons, with strong agreement between AI evaluations and expert human assessments. AI effectively automated skill assessment through real-time data capture and retrospective video analysis. Heterogeneity in study designs and the lack of standardized reporting metrics limited direct comparisons. AI-based laparoscopic skill assessment provides an objective, scalable alternative to traditional evaluations with distinguishing ability and automated skill assessment. Real-time AI feedback may accelerate skill acquisition, limit human bias, and improve training efficiency. Standardization of AI assessment metrics is needed to validate and integrate AI into surgical educational programs.

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